

Question 1: Pull and Run a Nginx Container

Objective: Pull an official image and run a container.

Task:

1. Pull the official `nginx` image from Docker Hub.
2. Run a container named `web-server` from the `nginx` image in detached mode.
3. Verify the container is running.
4. Open your browser or use `curl` to test that the server works.

Hints:

- Use `docker pull nginx`
 - Use `docker run -d --name web-server nginx`
 - Check with `docker ps`
 - Get the container's IP or use `localhost` (if port mapped) – *for simplicity, just check logs or inspect.*
-

Question 2: Access the Shell of a Running Container

Objective: Execute commands inside a running container.

Task:

1. Run an `ubuntu` container in interactive mode (so it stays alive).
2. Inside the container, create a file named `hello.txt` with the text "Docker is fun!".
3. Exit the container without stopping it (detach from it).
4. Re-attach to the same container and confirm the file exists.

Hints:

- Run with `docker run -it --name my-ubuntu ubuntu bash`
- Use `echo "Docker is fun!" > hello.txt`
- Detach with `Ctrl+P, Ctrl+Q` (not `exit`).
- Re-attach with `docker attach my-ubuntu`
- Use `cat hello.txt` to confirm.

Question 3: Stop, Start, and Remove a Container

Objective: Manage container lifecycle.

Task:

1. Run a container from the `alpine` image with the command `ping 8.8.8.8` (this will run forever).
2. Let it run for 10 seconds, then **stop** it gracefully.
3. **Start** the same container again.
4. **Stop** it once more, then **remove** it.
5. Verify it no longer exists.

Hints:

- Use `docker run -d --name pinger alpine ping 8.8.8.8`
 - Stop: `docker stop pinger`
 - Start: `docker start pinger`
 - Remove: `docker rm pinger` (container must be stopped)
 - Check with `docker ps -a`
-

Question 4: Clean Up – Remove All Stopped Containers

Objective: Use a single command to clean up.

Task:

1. Create **two** stopped containers (you can run and stop any images).
2. Use a single Docker command to remove **all** stopped containers.
3. Confirm that only running containers remain.

Hints:

- Run a container with `docker run --name temp1 alpine echo "done"` – it exits immediately.
- Run another: `docker run --name temp2 alpine echo "done"`.
- Use `docker container prune` OR `docker rm $(docker ps -aq)`.
- Check with `docker ps -a` to confirm.

Bonus (Optional Challenge): Port Mapping

Task: Run a `nginx` container and map port `8080` on your host to port `80` inside the container. Then access `http://localhost:8080` in your browser.

Hint:

- Use `-p 8080:80` with `docker run`.